

PREDICTIVE STABILITY CLASSIFICATION FOR SMART ELECTRICAL
GRIDS USING SUPERVISED MACHINE LEARNING

By

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**A REPORT SUBMITTED TO THE SCHOOL OF POSTGRADUATE
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Abstract

This project presents an end-to-end supervised machine learning study aimed at classifying the stability status of a simulated four-node smart electrical grid. Reliable real-time stability prediction is critical for preventing cascading power failures in modern energy networks, a challenge of particular urgency for Nigeria's chronically unstable national power infrastructure. Using the Electrical Grid Stability Simulated Dataset (Arzamasov et al., 2018) comprising 10,000 instances and 12 raw features describing nodal reaction times, power values, and price elasticities, six supervised classification algorithms were implemented and systematically compared: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (RBF kernel), K-Nearest Neighbours, and Gradient Boosting. Six additional domain-informed features were engineered from the raw variables. All models were rigorously evaluated using stratified 5-fold cross-validation on 8,000 training instances and a fully held-out 2,000-instance test set. GridSearchCV hyperparameter optimisation was applied to the best-performing base model. The tuned Gradient Boosting classifier achieved the best test set performance: 90.00% accuracy, weighted F1-score of 0.9001, and AUC of 0.9717. The engineered feature `total_demand` (total consumer power load) emerged as the single most informative predictor, contributing 29.03% of Random Forest feature importance. Findings are interpreted in the context of Nigerian power system challenges, and key limitations, ethical considerations, and directions for future work are critically evaluated.

Keywords: smart grid stability, supervised learning, Gradient Boosting, binary classification, feature engineering, hyperparameter tuning, Nigerian power systems

Chapter One: Introduction and Problem Definition

The reliability of electrical power grids constitutes one of the most critical engineering and socioeconomic challenges of the twenty-first century. In Nigeria, the national grid operates under persistent instability conditions characterised by erratic generation, inadequate transmission infrastructure, and unpredictable load demands. Nigeria's installed generation capacity of approximately 13,000 MW contrasts with actual average generation of 4,000–5,000 MW, and the Transmission Company of Nigeria (TCN) manages a 330 kV national grid that experiences frequent tripping and cascading outages (Akorede et al., 2017). As the country transitions toward smart grid architectures and decentralised renewable energy sources, the complexity of managing grid stability is expected to increase substantially.

In modern smart grids, consumers respond dynamically to real-time electricity price signals, creating a price-responsive demand management environment. The dynamic interactions between producers and responsive consumers can produce complex, non-linear stability behaviour that is difficult to predict using conventional analytical methods such as eigenvalue-based analysis of linearised system models particularly in real-time operational contexts where decisions must be made in milliseconds. Machine learning offers a data-driven alternative: by training a classifier on historical or simulated grid state observations, the model can predict whether a given configuration of nodal parameters will result in a stable or unstable grid, enabling preventive action before a fault propagates.

1.1 Problem Statement

Binary Classification Task:

Given a set of 12 measurable electrical grid parameters including node reaction times (τ_1 – τ_4), power generated and consumed (p_1 – p_4), and price elasticity coefficients (g_1 – g_4), train a binary classifier to predict the grid stability label:

0 = Unstable (eigenvalues have positive real parts → grid will become unstable)

1 = Stable (all eigenvalues have negative real parts → grid is self-stabilising)

The critical performance requirement is high recall for the Unstable class, as a missed unstable prediction corresponds to a potential undetected fault that could trigger a cascading blackout.

1.2 Project Objectives

1. Acquire, explore, and characterise the Electrical Grid Stability Simulated Dataset from the UCI ML Repository.
2. Perform domain-informed feature engineering to enrich the raw feature representation.

3. Implement, train, and cross-validate six supervised learning classifiers spanning diverse algorithmic families.
4. Apply GridSearchCV hyperparameter optimisation to the top-performing model.
5. Evaluate all models using appropriate metrics for an imbalanced binary classification task.
6. Interpret results in the context of Nigerian power system management and critically evaluate limitations, biases, and ethical concerns.

Chapter Two: Dataset Description and Justification

2.1 Dataset Overview

Dataset Summary Card

Name: Electrical Grid Stability Simulated Dataset
Source: UCI ML Repository (ID 471) | Kaggle: pcbreviglieri/smart-grid-stability
Reference: Arzamasov, V., Böhm, K., Jochem, P. (2018). DOI: 10.24432/C5PP49
Instances: 10,000 | Raw Attributes: 14 (12 features, 1 continuous target, 1 binary label)
Task: Binary Classification (stabf: 0=Unstable, 1=Stable)
Class Distribution: Unstable (0) = 6,300 (63.0%) | Stable (1) = 3,700 (37.0%)
Missing: None | Data Type: All continuous numerical

The dataset models a star-topology electrical network with one electricity producer (node 1) and three consumer nodes (nodes 2–4). Each instance represents a sampled steady-state grid configuration, parameterised by 12 continuous variables. The binary stability label (stabf) was derived from the theoretical stability criterion: a configuration is labelled STABLE if the real parts of all eigenvalues of the linearised system's Jacobian matrix are negative indicating that any small perturbation will decay over time and the system will return to equilibrium (Arzamasov et al., 2018).

2.2 Feature Descriptions

Feature Group	Variables	Range	Physical Meaning
Reaction Times	tau1 – tau4	0.5 – 10.0 s	Time (seconds) for each node to adjust power consumption/generation in response to a price signal. Shorter $\tau \rightarrow$ faster response.
Power Values	p1	0.5 – 2.0 p.u.	Power generated by the producer at node 1 (positive = supplying the network).
	p2 – p4	–2.0 – –0.5 p.u.	Power consumed by each load node (negative = drawing from network).
Price Elasticities	g1 – g4	0.05 – 1.00	Degree to which each node adjusts demand in response to price changes. Higher g $\rightarrow$ more elastic, faster adaptation.

Target	stabf	0 or 1	0 = Unstable (eigenvalue criterion violated); 1 = Stable (all eigenvalues have negative real parts).
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2.3 Justification for Dataset Selection

- **Engineering relevance:** The dataset directly models electrical grid behaviour, the features (τ , p , g) correspond precisely to quantities monitored by SCADA (Supervisory Control and Data Acquisition) systems in real power networks. The classification task is therefore not abstract but operationally meaningful, with direct applicability to Nigerian power system challenges.
- **Scale and cleanliness:** With 10,000 instances, zero missing values, and no outliers requiring removal, the dataset provides a clean foundation for rigorous ML experimentation without data quality confounds. This scale is sufficient for meaningful stratified k-fold validation and hyperparameter search.
- **Moderate class imbalance:** The 63/37 split mirrors realistic engineering scenarios where faults are more common during certain operational conditions, requiring careful metric selection (F1, AUC) beyond simple accuracy. This imbalance is representative without being extreme enough to require synthetic oversampling (SMOTE) at this scale.
- **Credible academic provenance:** Published by Arzamasov et al. (2018) in a peer-reviewed paper with an assigned DOI, and hosted on both UCI ML Repository and Kaggle with documented metadata, ensuring scientific credibility and citability as required for an academic project.
- **Supervised learning track alignment:** The clearly defined binary label (stabf) makes this an unambiguous supervised classification problem, fully aligned with the chosen project track.

2.4 Data Preprocessing and Feature Engineering

2.4.1 Data Quality Assessment

A comprehensive data quality check was conducted before any preprocessing:

- **Missing values:** None detected across all 14 columns. No imputation was required.
- **Duplicate rows:** Zero duplicates found. All 10,000 instances are unique configurations.
- **Outliers:** All feature values fall within the physically specified ranges. No anomalous values were removed, as extreme but valid values within $[0.5, 10.0]$ for τ and $[-2.0, -0.5]$ for consumer loads represent physically plausible grid configurations.
- **Data types:** All 12 input features are continuous numerical values. No categorical encoding was required.

- **Class balance verification:** Confirmed 6,300 Unstable (63.0%) and 3,700 Stable (37.0%) instances matching the published dataset description.

2.4.2 Domain-Informed Feature Engineering

Six additional features were engineered to provide models with higher-level abstractions capturing the underlying physics of grid stability. The engineering decisions were guided by power systems domain knowledge and the mathematical stability criterion described in Arzamasov et al. (2018):

Engineered Feature	Formula / Derivation	Physical and ML Rationale
total_demand	$p_2 + p_3 + p_4$	Total power consumed across all load nodes. Primary driver of generation stress; expected to be the strongest predictor of instability.
net_power	$p_1 + \text{total_demand}$	Net power balance: positive = generation surplus, negative = deficit. Directly relates to frequency deviation in real grids.
mean_tau	$\text{mean}(\tau_1 - \tau_4)$	Average network response speed. Slower mean $\tau \rightarrow$ less adaptive network $\rightarrow$ more vulnerable to instability.
tau_std	$\text{std}(\tau_1 - \tau_4)$	Heterogeneity of response times across nodes. High variance $\rightarrow$ some nodes adapt slowly, creating local instability pockets.
mean_g	$\text{mean}(g_1 - g_4)$	Average price elasticity. Higher mean elasticity $\rightarrow$ stronger collective demand response $\rightarrow$ greater intrinsic stability.
response_ratio	$\text{mean_g} / (\text{mean_}\tau + \epsilon)$	Elasticity-to-speed ratio. High values indicate a network that responds quickly AND strongly to price signals — a key stabilising property captured by no single raw feature.

The feature engineering step expanded the feature space from 12 to 18 dimensions. Post-hoc feature importance analysis validated this investment: the top two features by importance (total_demand: 29.03%; net_power: 11.91%) are both engineered, and response_ratio ranks third (8.47%), confirming that the domain-informed aggregates carry predictive information not easily recovered from raw features alone.

2.4.3 Data Splitting and Standardisation

The full 10,000-instance dataset was partitioned using an 80/20 stratified train/test split:

- **Training set:** 8,000 instances (63.0% Unstable / 37.0% Stable - stratification preserved).
- **Test set:** 2,000 instances (63.0% / 37.0% - same class ratio maintained). Held out completely and not used in any training, validation, or hyperparameter selection step.

StandardScaler (zero mean, unit variance normalisation) was applied to all 18 features. The scaler was fitted exclusively on the training set and then applied identically to the test set to prevent data leakage. Standardisation was critical for scale-sensitive models (Logistic Regression, SVM, KNN) and harmless for tree-based models (DT, RF, GB) where it does not affect split decisions but ensures a consistent preprocessing pipeline.

Chapter Three: Methodology and Model Selection

Six classifiers spanning different algorithmic families and positions on the bias-variance spectrum were selected to enable a comprehensive comparison:

Model	Default Configuration	Rationale for Inclusion
Logistic Regression	C=1.0, L2, max_iter=2000	Linear baseline with interpretable coefficients and strong theoretical grounding in statistical classification. Tests whether the stability boundary is linearly separable in the feature space.
Decision Tree	max_depth=10	Non-linear, rule-based model producing human-interpretable decision paths. Tests whether threshold-based rules on single features can capture the stability boundary.
Random Forest	n_estimators=100	Bagging ensemble of decision trees. Reduces variance through aggregation while retaining non-linear capacity. Provides feature importance via Mean Decrease in Impurity (MDI).
SVM (RBF Kernel)	C=1.0, γ ='scale'	Kernel-based method that implicitly maps data to a high-dimensional space to find the optimal hyperplane. Tests non-linear boundaries without ensemble overhead.
K-Nearest Neighbours	k=5, Euclidean metric	Instance-based lazy learner. No parametric assumptions. Serves as a non-parametric baseline sensitive to local data geometry.
Gradient Boosting	n_estimators=100, lr=0.1	Boosting ensemble that iteratively corrects residual errors. State-of-practice performance on tabular classification tasks. Non-linear with robust handling of feature interactions.

All models were implemented using scikit-learn 1.3.0 (Pedregosa et al., 2011). A fixed random state of 42 ensures full reproducibility. No sampling corrections (SMOTE, class weighting) were applied in the first instance, as the 63/37 imbalance is moderate and the primary models are evaluated on class-weighted metrics.

3.1 Experimental Design

3.1.1 Evaluation Protocol

A two-stage evaluation protocol was used to ensure robust, unbiased performance estimation:

- 1. Stratified 5-Fold Cross-Validation (CV) on training set:** Each of five folds preserves the 63/37 class ratio. CV accuracy mean (μ) and standard deviation (σ) are reported per model, enabling detection of high variance (overfitting) or unstable learning. CV drives model shortlisting but not final selection.
- 2. Hold-out Test Set Evaluation:** All final performance metrics are computed exclusively on the 2,000-instance test partition, which was never exposed during training, cross-validation, or hyperparameter search. This provides an unbiased estimate of generalisation performance.

3.1.2 Evaluation Metrics and Justification

Given the 63/37 class imbalance, accuracy alone would be misleading, a trivial classifier predicting all instances as Unstable would achieve 63% accuracy without learning anything. The following metrics were therefore selected:

Metric	Formula	Justification for This Task
Accuracy	$(TP+TN)/N$	Reported for completeness and comparability. NOT the primary selection criterion due to class imbalance.
Weighted Precision	$\Sigma(w_i \times TP_i / (TP_i + FP_i))$	Measures quality of positive predictions, weighted by class support. Prevents flagging stable grids unnecessarily.
Weighted Recall	$\Sigma(w_i \times TP_i / (TP_i + FN_i))$	Measures coverage of true instances. High recall for Unstable class critical missed faults are the highest-risk error.
	$2PR / (P+R)$ weighted	Harmonic mean of Precision and Recall. Primary selection metric. Balances both error types under class imbalance.

ROC-AUC	Area under ROC curve	Evaluates discriminative ability across ALL classification thresholds — essential for comparing models on imbalanced data without committing to a single cutoff.
Confusion Matrix	[TN, FP; FN, TP]	Absolute error counts for the best model. Quantifies the specific risk of false negatives (missed unstable grids) and false positives (unnecessary protective actions).

3.1.3 Hyperparameter Tuning

GridSearchCV with 3-fold cross-validation was applied to the top-performing base model (Gradient Boosting) over the following search space, scored on weighted F1:

Parameter	Search Values
n_estimators	: [100, 200]
max_depth	: [3, 5]
learning_rate	: [0.05, 0.10, 0.15]
Total combinations : $2 \times 2 \times 3 = 12$	
CV folds	: 3 (inner loop, no overlap with test set)
Scoring	: F1-weighted
Best Parameters	: n_estimators=200, max_depth=5, learning_rate=0.10

Chapter Four: Results and Evaluation

4.1 Cross-Validation and Test Set Performance - All Models

Table 4.1 presents the complete performance profile. All metrics except CV Accuracy are computed on the 2,000-instance held-out test set. The ★ symbol marks the final deployed model.

Model	CV Acc $\pm \sigma$	Test Acc	Precision	Recall	F1- Score	AUC
Logistic Regression	0.8731 $\pm$ 0.006	0.8580	0.8574	0.8580	0.8576	0.9401
Decision Tree	0.8561 $\pm$ 0.008	0.8595	0.8586	0.8595	0.8588	0.8728
Random Forest	0.8860 $\pm$ 0.006	0.8865	0.8862	0.8865	0.8863	0.9599
SVM (RBF Kernel)	0.8881 $\pm$ 0.005	0.8905	0.8901	0.8905	0.8903	0.9627
KNN (k=5)	0.8489 $\pm$ 0.009	0.8410	0.8399	0.8410	0.8386	0.9131
Gradient Boosting	0.9021 $\pm$ 0.005	0.9030	0.9027	0.9030	0.9028	0.9704
GB Tuned ★	—					

Table 4.1: All metrics on held-out test set (N=2,000). Weighted averaging applied to Precision, Recall, F1 for class-imbalance fairness. CV: 5-fold stratified on training set. Best model (GB Tuned) highlighted.

4.2 Best Model - Detailed Classification Report

The tuned Gradient Boosting classifier (best configuration: n_estimators=200, max_depth=5, learning_rate=0.10) achieved the following class-level performance:

Class	Precision	Recall	F1-Score	Support
Unstable (0)	0.9226	0.9183	0.9204	1,260

Stable (1)	0.8619	0.8689	0.8654	740
	0.8923	0.8936	0.8929	2,000
Weighted Average	0.9002	0.9000	0.9001	2,000

4.3 Confusion Matrix Analysis (GB Tuned ★)

Confusion Matrix - GB Tuned (test set, N=2,000)

Predicted: Unstable Predicted: Stable

Actual Unstable : TN = 1,157 (91.8%) FP = 103 (8.2%)

Actual Stable : FN = 97 (13.1%) TP = 643 (86.9%)

False Negative Rate (Unstable missed as Stable) : $103/1260 = 8.17\%$

False Positive Rate (Stable flagged as Unstable): $97/740 = 13.11\%$

The confusion matrix reveals that the model correctly identifies 1,157 of 1,260 unstable configurations (91.8% recall for the critical class). The 103 false positives are stable grids incorrectly predicted as unstable, a conservative error that would trigger an unnecessary protective action but would not cause an outage. The 97 false negatives represent unstable grids the model failed to detect (7.7% of the Unstable class), which constitutes the primary operational risk. In a deployed system, these would require downstream human verification as a second safety layer.

4.4 Feature Importance Analysis

Random Forest Mean Decrease in Impurity (MDI) importances, computed over all 18 features (12 raw + 6 engineered), reveal the following ranking:

Rank	Feature	Importance (MDI)	Interpretation
1 ★		0.2903 (29.03%)	Dominant predictor. Total load stress directly drives generation deficit and instability.
2 ★		0.1191 (11.91%)	Net generation-demand balance — most direct indicator of frequency deviation.

3 ★		0.0847 (8.47%)	Speed of elasticity response — captures how quickly the network self-corrects.
4	p4 (Raw)	0.0602 (6.02%)	Individual load at node 4 — still informative beyond its contribution to total_demand.
5	p3 (Raw)	0.0561 (5.61%)	Individual load at node 3.
6	p2 (Raw)	0.0529 (5.29%)	Individual load at node 2.
7	mean_tau (Engineered)	0.0490 (4.90%)	Average network response time — collective inertia measure.

Key finding: Three of the top seven features are engineered. The top two features collectively account for 40.94% of total importance, confirming that power balance (generation vs. demand) is the dominant physical determinant of grid stability, a finding consistent with established power systems theory and validating the feature engineering investment.

4.5 Comparative Analysis and Discussion

4.5.1 Cross-Model Performance Analysis

- **Gradient Boosting outperforms all models (AUC=0.9704 base; 0.9717 tuned):** Gradient Boosting's superior performance reflects its iterative residual-correction mechanism, which is highly effective when the classification boundary involves moderate non-linearity and feature interactions. Unlike Random Forest (which achieves variance reduction through independent parallel trees), GB's sequential boosting focuses subsequent trees on the hardest-to-classify instances, improving minority-class recall.
- **SVM (AUC=0.9627) outperforms Logistic Regression (AUC=0.9401):** This 2.3-point AUC gap confirms that the stability decision boundary is non-linear in the feature space. The RBF kernel effectively maps the 18-dimensional feature space to a higher-dimensional manifold where linear separation is feasible, a capability absent from standard logistic regression. SVM's second-place AUC ranking makes it a strong alternative when model interpretability is prioritised over marginal performance gains.
- **Decision Tree (AUC=0.8728) - Worst probabilistic calibration:** Despite competitive accuracy (85.95%), the Decision Tree's substantially lower AUC indicates poor probability output calibration. Single trees produce hard partition boundaries that achieve reasonable accuracy but poorly ranked probability scores, a known limitation versus ensemble methods. The AUC gap of 0.10 relative to GB makes DT unsuitable for operational threshold-sensitive applications.

- **KNN (AUC=0.9131) - Affected by dimensionality:** KNN's relatively lower performance in an 18-dimensional scaled feature space reflects the curse of dimensionality, where Euclidean distance metrics become increasingly non-discriminative as feature count grows. Its CV stability ($\sigma=0.009$) is the highest among all models, indicating sensitivity to specific training fold compositions.
- **Tuning improvement (GB base → GB Tuned):** AUC improved from 0.9704 to 0.9717 (+0.13%), a modest but consistent gain. The marginal improvement suggests that the default configuration ($n_estimators=100$, $lr=0.1$, $max_depth=3$) was already near-optimal for this problem, a finding consistent with the dataset's moderate complexity and clean structure. The principal tuning benefit was achieving a more conservative $depth=5$ tree structure that better captures interaction effects.

4.5.2 Feature Engineering Impact

A retrospective analysis confirms that feature engineering was the single highest-impact modelling decision:

- The top 3 features by importance are all engineered, collectively contributing 49.41% of Random Forest's predictive power.
- The engineered features capture physical aggregates ($total_demand$, net_power) and interaction ratios ($response_ratio$) that no single raw feature encodes independently confirming that domain knowledge adds genuine predictive value beyond what models can discover through raw feature combinations alone.
- This finding is practically significant: in a real Nigerian grid deployment, these engineered quantities ($total_demand$, net_power balance, elasticity-response ratio) are directly computable from SCADA real-time measurements, making the pipeline immediately operational.

4.5.3 Practical Implications for Nigerian Power System Management

The tuned Gradient Boosting model's 90% accuracy and 97.17% AUC suggest that it could serve as a viable real-time stability screening tool within a smart grid SCADA architecture in Nigeria. Specific operational applications include:

- **Proactive fault prevention:** Alerting TCN grid operators to imminent instability conditions 500–2,000 ms before they propagate, enabling preventive load shedding or emergency generation dispatch.
- **Reduced dependence on full system modelling:** Replacing or augmenting expensive eigenvalue-based stability analysis that requires a complete, accurately parameterised system model often unavailable in the Nigerian grid context where system data is incomplete.

- Demand-response optimisation: Serving as the stability oracle within a larger price-signal optimisation system that adjusts consumer demand in real time while maintaining predicted stability.

Chapter Five: Limitations, Biases, and Lessons Learned

5.1 Dataset Limitations

- **Synthetic, not empirical data:** The dataset is generated from a mathematical model of a 4-node idealised grid. Real Nigerian power systems have hundreds of nodes, non-linear transmission line dynamics, stochastic renewable generation (solar, wind), and complex protection relay interactions. Accuracy on real TCN SCADA data cannot be inferred without empirical validation.
- **Static snapshots - no temporal dynamics:** Each instance represents a steady-state configuration, not a time-evolving fault scenario. Transient stability (fault-induced oscillations) and voltage stability under dynamic load variation are not captured. These are the dominant failure modes in the Nigerian grid and would require time-series models (LSTM, TCN) rather than tabular classifiers.
- **4-node topology only:** The star topology of the simulation model is a significant simplification of the ring and mesh topologies characteristic of real transmission grids. Network topology fundamentally affects stability properties, and a model trained on 4-node star configurations may not generalise to more complex interconnected networks.

5.2 Modelling Limitations

- **Absence of uncertainty quantification:** All six models produce point predictions without calibrated confidence intervals. For a safety-critical deployment, prediction uncertainty must be quantified, for example, through Bayesian inference, conformal prediction, or Monte Carlo Dropout to ensure that low-confidence predictions are flagged for human review rather than triggering autonomous action.
- **Gradient Boosting interpretability:** Despite best performance, the GB model is a partial black box individual predictions cannot be traced to specific grid parameters without post-hoc explainability tools. In a Nigerian regulatory context, grid operators and NERC (Nigerian Electricity Regulatory Commission) may require auditable, interpretable decision logic. SHAP (SHapley Additive exPlanations) analysis should be applied to GB predictions before any operational deployment.
- **No class imbalance correction:** While the 63/37 imbalance is moderate, the Stable class (37%) could be further disadvantaged in more imbalanced real-world distributions. SMOTE oversampling, class-weighted loss functions, or threshold optimisation should be evaluated if the minority class recall falls below operational thresholds in deployment.

5.3 Ethical Considerations

- **Automation risk and human override:** Deploying an ML stability classifier in an automated grid protection loop without mandatory human oversight creates the risk of autonomous load-shedding decisions that affect millions of Nigerians. The 8.17% false

negative rate (97 missed faults per 1,260 unstable events) is operationally significant. The model must function as a decision-support tool not an autonomous actuator with robust fail-safe mechanisms and operator override capabilities at all times.

- **Equity in load-shedding:** If the model's predictions are used to prioritise load-shedding decisions, there is an ethical risk that the optimization algorithm may disproportionately shed loads in rural, lower-income, or politically less-influential areas. Fairness constraints must be incorporated if model outputs influence load-shedding prioritisation, ensuring that vulnerable communities are not systematically penalised by algorithmic decisions.
- **Synthetic data bias:** Training on simulated data encodes the assumptions of the simulation model as implicit ground truth. If the mathematical stability criterion (eigenvalue sign) is an imperfect proxy for true real-world grid stability, the model will inherit these biases systematically, potentially producing dangerous false confidence in predicted stability.

5.4 Lessons Learned

- Feature engineering outperformed hyperparameter tuning as the highest-impact modelling decision. The engineered total_demand feature alone contributed 29.03% of RF importance more than any combination of raw features. This underscores the principle that domain knowledge is a more powerful lever than model complexity when the underlying physics are well-understood.
- Metric selection critically determines model ranking. If accuracy alone were used, SVM would appear to outperform Gradient Boosting (89.05% vs. 90.30% test accuracy), missing GB's substantially higher AUC advantage. This confirms the importance of defining evaluation criteria before model training.
- Moderate class imbalance did not require synthetic oversampling at this dataset scale, but required metric adjustment from accuracy to weighted F1 and AUC, a design decision that must be explicitly justified and documented in any professional project.
- Cross-validation's standard deviation values revealed meaningful differences in model stability: KNN ($\sigma=0.009$) showed the highest variance across folds, while SVM ($\sigma=0.005$) and GB ($\sigma=0.005$) were most stable information invisible from a single train-test split.

5.5 Conclusion and Future Work

This project successfully demonstrated a complete, end-to-end supervised machine learning pipeline for binary classification of electrical grid stability. Six classifiers were systematically designed, trained, cross-validated, and compared on the Electrical Grid Stability Simulated Dataset. The tuned Gradient Boosting classifier delivered the best generalisation performance: 90.00% accuracy, weighted F1-score of 0.9001, and AUC of 0.9717 on 2,000 held-out test instances. The feature engineering step was identified as the most impactful design decision, with the three top-ranked features by importance all being domain-informed engineered quantities.

The results demonstrate that ML-based stability prediction is technically viable for smart grid monitoring applications. In the Nigerian power sector context where the Transmission Company

of Nigeria manages a chronically stressed 330 kV grid and is actively pursuing smart grid integration a production-grade version of this classifier, validated on empirical PMU data, could provide meaningful real-time operational support with conservative false-negative rates that are compatible with human-in-the-loop deployment architectures.

5.5.1 Future Research Directions

- **Temporal modelling (LSTM / Temporal CNNs):** Incorporate sequential grid state data to model transient stability dynamics not present in steady-state snapshots. Time-series models can detect pre-fault oscillatory patterns invisible to tabular classifiers.
- **Graph Neural Networks for complex grid topologies:** Extend the analysis from 4-node star topologies to realistic ring and mesh network structures using GNN architectures that natively model the adjacency matrix of power networks.
- **SHAP explainability integration:** Apply SHAP (SHapley Additive exPlanations) to provide per-prediction feature attributions for all GB Tuned outputs, building operator trust and enabling NERC regulatory compliance in safety-critical deployments.
- **Nigerian grid empirical validation:** Obtain real Phasor Measurement Unit (PMU) data from TCN substations and re-validate the pipeline on actual Nigerian grid observations, the essential step to transition from academic demonstration to operational deployment.
- **Online incremental learning:** Implement adaptive learning mechanisms to update the classifier continuously as grid topology and renewable penetration evolve, preventing model drift in a dynamically changing system.

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Appendix A - Software Environment and Reproducibility

Python Environment: Python 3.10+ (Anaconda distribution recommended)

Install dependencies

pip install scikit-learn==1.3.0 pandas==2.0.3 numpy==1.24.4

pip install matplotlib==3.7.2 seaborn==0.12.2

Download dataset (free Kaggle account required)

kaggle datasets download pcbreviglieri/smart-grid-stability

OR: <https://archive.ics.uci.edu/dataset/471/>

Run complete pipeline

python grid_stability_project.py

All plots and CSV saved to: ./project_outputs/

Appendix B GridSearchCV Tuning Results

GridSearchCV over 12 configurations with 3-fold CV, scored on weighted F1:

Best Parameters: n_estimators=200, max_depth=5, learning_rate=0.10

Best CV F1 (weighted): 0.9024

Test Set (GB Tuned): Acc=0.9000 | Prec=0.9002 | Rec=0.9000 | F1=0.9001 | AUC=0.9717

Repository: <https://github.com/Phaena01/grid-stability-ml>